

# THERMOCHEMICAL MODELING OF A PHENOLIC LINER IN A SOLID ROCKET MOTOR USING ANSYS MECHANICAL

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**Purpose of this document.** *This report first describes the transient thermal model currently being run, limited to volumetric pyrolysis of the inner phenolic layer **phe0**. It then proposes an extension, still entirely within ANSYS Mechanical, to represent outgassing and discrete recession through element removal. Fluent and propellant combustion are outside the scope of this study. Values marked provisional must be replaced or validated using measurements of the material actually manufactured before any design or safety decision is made.*

## 1 Context: Solid Rocket Motor and Casing Layup

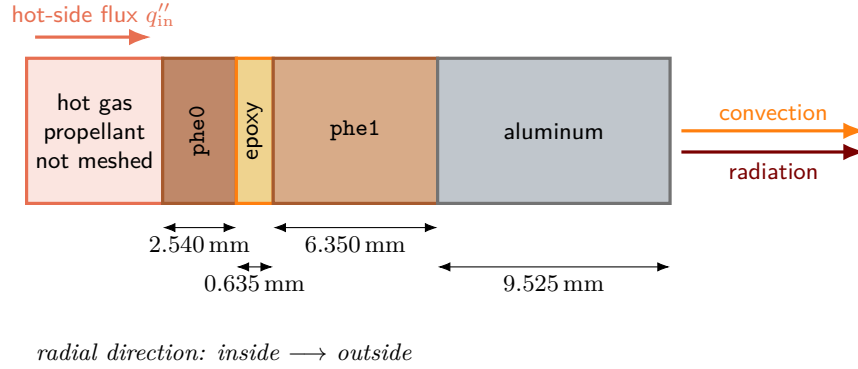
### 1.1 Physical Scope

The system under study is a solid-propellant rocket motor whose casing is modeled as four contacting coaxial tubes: an inner phenolic layer, a thin epoxy layer, a second phenolic layer, and an aluminum shell. The geometry file contains the bodies **Phenolic0**, **Epoxy**, **Phenolic1**, and **Alu**, which correspond respectively to the Mechanical named selections **phe0**, **epo**, **phe1**, and **alu**. Mechanical nevertheless meshes only an axially truncated portion of this baseline model: the radial layup is preserved, but the reduced height simplifies the mesh and keeps the solution time reasonable. The dimensions below describe the complete motor, so extensive results must be scaled before extrapolation to the full motor.

The propellant is not included in the geometry. Its effect is represented by a prescribed gas temperature and convection coefficient on the inner surface of **phe0**. It should also be noted that ammonium perchlorate (AP) is the *oxidizer* in a composite propellant, not a fuel by itself. The composition of the fuel binder, any aluminum fraction, chamber pressure, and combustion history are not documented in the present model. Consequently, a wall heat flux cannot be derived from first principles: the current hot-side boundary condition remains an input that must be calibrated against a motor firing or supplied by a separate internal-flow model.

## 1.2 Dimensions Extracted from the 3D Model

The AP214 STEP file uses meters. Its cylindrical surfaces define the interface radii listed in Table 1. The model is a straight cylinder of length  $L = 1.8288$  m, exactly 72 in; its bore diameter is 266.7 mm (10.5 in), and its outside diameter is 304.8 mm (12 in). The layers have a combined thickness of 19.05 mm.



**Figure 1.** Radial layup extracted from the 3D model. Layer widths are schematic; the stated dimensions are exact.

| Body | Inner radius<br>(mm) | Outer radius<br>(mm) | Thickness<br>(mm) | Estimated initial mass<br>(kg) |
|------|----------------------|----------------------|-------------------|--------------------------------|
| phe0 | 133.350              | 135.890              | 2.540             | 4.911                          |
| epo  | 135.890              | 136.525              | 0.635             | 1.143                          |
| phe1 | 136.525              | 142.875              | 6.350             | 12.742                         |
| alu  | 142.875              | 152.400              | 9.525             | 43.629                         |

**Table 1.** Motor geometry. Masses are calculated using the model’s initial densities and exclude end closures, seals, interfaces, and propellant.

The hot-side cylindrical area is  $2\pi r_i L = 1.5323 \text{ m}^2$ , and the outside area is  $2\pi r_o L = 1.7512 \text{ m}^2$ . These values will be useful for solver-independent energy and mass balances.

## 1.3 Materials and Interfaces

The inner body **phe0** uses the user material PHENOLIC\_PYROLYSIS; **phe1** uses PHENOLIC\_VIRGIN; the epoxy and aluminum bodies use the project’s generic materials. All three concentric interfaces are program-controlled bonded thermal contacts. This configuration represents perfect adhesion with neither debonding nor thermal contact resistance. It is appropriate for an initial study, but it does not investigate delamination, epoxy charring, or the formation of a gas-filled gap.

The initial density is  $1250 \text{ kg m}^{-3}$  for virgin phenolic,  $1150 \text{ kg m}^{-3}$  for epoxy, and  $2700 \text{ kg m}^{-3}$  for aluminum. The aluminum alloy and exact epoxy grade are unspecified. Their associated properties are therefore generic approximations, not qualified material data.

## 2 Simulation 1: Volumetric Pyrolysis of the First Phenolic Layer

### 2.1 Theory and Derivation

#### 2.1.1 Local Conservation of Energy

Consider a fixed material volume  $V$  of phenolic, bounded by  $\partial V$ , with no macroscopic motion of the solid. The first law of thermodynamics requires the rate of change of internal energy to equal the net heat input by conduction plus the volumetric heat generation:

$$\frac{d}{dt} \int_V \rho u \, dV = - \int_{\partial V} \mathbf{q} \cdot \mathbf{n} \, dS + \int_V \rho \dot{q}_m \, dV. \quad (1)$$

Here,  $u$  is the specific internal energy,  $\rho$  is the density of the condensed medium,  $\dot{q}_m$  is a specific heat source, and  $\mathbf{q}$  is the heat flux. Applying the divergence theorem to the surface integral and noting that the volume is arbitrary gives

$$\frac{\partial(\rho u)}{\partial t} = -\nabla \cdot \mathbf{q} + \rho \dot{q}_m. \quad (2)$$

Using Fourier's law,  $\mathbf{q} = -k\nabla T$ , and identifying  $\partial u / \partial T$  with  $c_p$  in the Mechanical thermal formulation yields the form used in the model:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \rho \dot{q}_m. \quad (3)$$

Pyrolysis is endothermic. If  $\Delta H_p > 0$  is the enthalpy absorbed per kilogram of converted material and  $\alpha$  is the degree of conversion, the specific heat source is

$$\dot{q}_m = -\Delta H_p \dot{\alpha}, \quad (4)$$

and therefore

$$\boxed{\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) - \rho \Delta H_p \dot{\alpha}.} \quad (5)$$

This first simulation does not transport the enthalpy of the pyrolysis gases and does not move the hot boundary. Equation (5) is therefore a conduction-reaction equation, not a complete ablation model.

#### 2.1.2 Conversion Variable and Mass Loss

At a given temperature, the model interpolates between the virgin state  $v$  and the char residue  $c$ . The apparent density of the solid is written as

$$\rho_s(T, \alpha) = (1 - \alpha)\rho_v(T) + \alpha\rho_c(T). \quad (6)$$

The equivalent definition of conversion is

$$\alpha = \frac{\rho_v - \rho_s}{\rho_v - \rho_c}, \quad 0 \leq \alpha \leq 1. \quad (7)$$

Thus,  $\alpha = 0$  denotes virgin material and  $\alpha = 1$  denotes the residual *char*. This distinction is essential:  $\alpha = 1$  means neither a void nor a fully consumed element. In this single-reaction model, the rate of volatile matter released per unit volume is

$$\dot{\omega}_g''' = (\rho_v - \rho_c)\dot{\alpha}. \quad (8)$$

The other properties are mixed linearly:

$$X(T, \alpha) = (1 - \alpha)X_v(T) + \alpha X_c(T), \quad X \in \{k, c_p, \rho\}. \quad (9)$$

This rule is simple and monotonic, but it is not a universal constitutive law for porous composites. Its validation requires measurements at intermediate conversion states or a sensitivity analysis.

#### 2.1.3 Arrhenius Kinetics and Integration over an Increment

The frequency of activated molecular events is represented by the Arrhenius rate constant

$$k_r(T) = A \exp\left(-\frac{E_a}{RT}\right), \quad (10)$$

where  $A$  is the pre-exponential factor,  $E_a$  is the activation energy, and  $R = 8.314462618 \text{ J mol}^{-1} \text{ K}^{-1}$ . Assuming that the remaining quantity of reactive sites is proportional to  $(1 - \alpha)^n$ ,

$$\boxed{\dot{\alpha} = k_r(T)(1 - \alpha)^n.} \quad (11)$$

During the current increment  $[t, t + \Delta t]$ , the code evaluates  $T$  at the midpoint temperature and treats  $k_r$  as constant. Let  $\beta = 1 - \alpha$ . Then  $\dot{\beta} = -k_r \beta^n$ . For  $n = 1$ ,

$$\int_{\beta_0}^{\beta_1} \frac{d\beta}{\beta} = -k_r \int_t^{t+\Delta t} dt, \quad (12)$$

hence

$$\beta_1 = \beta_0 \exp(-k_r \Delta t), \quad \boxed{\alpha_1 = 1 - (1 - \alpha_0) \exp(-k_r \Delta t)}. \quad (13)$$

For  $n \neq 1$ , integration gives

$$\frac{\beta_1^{1-n} - \beta_0^{1-n}}{1-n} = -k_r \Delta t, \quad (14)$$

or

$$\boxed{\beta_1 = [\beta_0^{1-n} - (1-n)k_r \Delta t]^{1/(1-n)}, \quad \alpha_1 = 1 - \beta_1.} \quad (15)$$

The bracketed term is bounded to prevent a nonphysical value. The stored rate is the average over the increment,  $\dot{\alpha} = (\alpha_1 - \alpha_0)/\Delta t$ . The code requests a time-step cutback if  $\Delta\alpha > 0.05$ , thereby limiting excessively abrupt numerical conversion. The *current increment* specifically means the substep attempted or accepted by the solver between two converged time points, not the complete seven-second load step.

### 2.1.4 Finite-Element Weak Form

Multiplying Equation (5) by a test function  $N_i$  and integrating the conduction term by parts gives

$$\int_V N_i \rho c_p \frac{\partial T}{\partial t} dV + \int_V \nabla N_i \cdot k \nabla T dV = - \int_{\partial V} N_i \mathbf{q} \cdot \mathbf{n} dS \quad (16)$$

$$- \int_V N_i \rho \Delta H_p \dot{\alpha} dV. \quad (17)$$

After applying the approximation  $T = \sum_j N_j T_j$ , the matrix form is

$$\mathbf{C}(T, \alpha) \dot{\mathbf{T}} + \mathbf{K}(T, \alpha) \mathbf{T} = \mathbf{f}_q - \mathbf{f}_p(T, \alpha). \quad (18)$$

The dependence of  $k$ ,  $c_p$ ,  $\rho$ , and  $\dot{\alpha}$  on temperature and history makes the problem nonlinear. `UserMatTh` is called at every integration point and returns the heat flux, its tangent, the heat capacity, the density, the specific heat source, and the state variables [1].

## 2.2 Current ANSYS Mechanical Configuration

### 2.2.1 Discretization, Contacts, and Boundary Conditions

All four bodies use SOLID279 three-dimensional thermal elements. The inspected export contains approximately 482 900 solid elements and 2 242 327 equations. Their distribution is given in Table 2. The eight layers through `phe0` yield an average radial thickness of only 0.318 mm, which is suitable for resolving a pyrolysis front. Even with the axial truncation described above, this radial resolution remains computationally expensive.

| Body | Elements | Mean radial layers | Material           |
|------|----------|--------------------|--------------------|
| phe0 | 315536   | 8                  | PHENOLIC_PYROLYSIS |
| epo  | 9516     | 1                  | generic epoxy      |
| phe1 | 59028    | 6                  | PHENOLIC_VIRGIN    |
| alu  | 98820    | 9                  | generic aluminum   |

**Table 2.** Mesh of the axially truncated domain exported by Mechanical.

The three contact pairs are bonded: `phe0_ext-epo_int`, `epo_ext-phe1_int`, and `phe1_ext-alu_int`. The initial temperature is  $T_0 = 295.15$  K. The inner face is subjected to

$$q''_{\text{in}} = h_{\text{in}}(T_g - T_s), \quad h_{\text{in}} = 1000 \text{ W m}^{-2} \text{ K}^{-1}, \quad T_g = 2504 \text{ K}. \quad (19)$$

This corresponds to  $2230.85^\circ \text{C}$  in the Mechanical interface. On the outside, the model combines natural convection with  $h_{\text{out}} = 8 \text{ W m}^{-2} \text{ K}^{-1}$  toward 295.15 K and gray-body radiation:

$$q''_{\text{rad}} = \varepsilon \sigma (T_s^4 - T_\infty^4), \quad \varepsilon_{\text{Al}} = 0.25. \quad (20)$$

The coefficients  $h_{\text{in}}$  and  $h_{\text{out}}$ , the gas temperature, and the emissivity are project assumptions, not outputs from a combustion calculation.

The study is a full 7 s transient thermal analysis with automatic time stepping:  $\Delta t_{\text{initial}} = 0.05$  s,  $\Delta t_{\text{min}} = 0.001$  s,  $\Delta t_{\text{max}} = 0.05$  s. The user routine adds its own criterion,  $\Delta\alpha \leq 0.05$ . State-variable results are requested at every substep through `OUTRES,SVAR,ALL`. The inspected configuration launches eight DMP processes; the routine uses no mutable global variables and is therefore also suitable for a future distributed run, provided that the DLL is accessible to every process.

### 2.2.2 Thermal Properties and Kinetic Parameters

Table 3 reproduces the property curves actually passed to `UserMatTh`. Mechanical uses degrees Celsius in this export, whereas the table is presented in kelvins. ANSYS interpolates the properties between the tabulated points; extrapolation beyond the final point should be avoided.

| $T$<br>(K) | $k_v$<br>(W m <sup>-1</sup> K <sup>-1</sup> ) | $k_c$ | $c_{p,v}$<br>(J kg <sup>-1</sup> K <sup>-1</sup> ) | $c_{p,c}$ | $\rho_v$<br>(kg m <sup>-3</sup> ) | $\rho_c$ |
|------------|-----------------------------------------------|-------|----------------------------------------------------|-----------|-----------------------------------|----------|
| 300        | 0.25                                          | 0.30  | 900                                                | 800       | 1250                              | 700      |
| 600        | 0.32                                          | 0.35  | 1100                                               | 900       | 1200                              | 650      |
| 900        | 0.36                                          | 0.40  | 1300                                               | 1000      | 1150                              | 600      |
| 1050       | 0.36                                          | 0.44  | 1300                                               | 1250      | 1150                              | 600      |
| 1200       | 0.36                                          | 0.49  | 1300                                               | 1550      | 1150                              | 600      |
| 1500       | 0.36                                          | 0.68  | 1300                                               | 1950      | 1150                              | 600      |
| 1700       | 0.36                                          | 0.81  | 1300                                               | 2060      | 1150                              | 600      |
| 2200       | 0.36                                          | 1.24  | 1300                                               | 2085      | 1150                              | 600      |
| 2750       | 0.36                                          | 1.73  | 1300                                               | 2090      | 1150                              | 600      |

**Table 3.** Currently encoded virgin/char property curves for the `PHENOLIC_PYROLYSIS` material. These are project engineering curves, not yet a certified characterization of the manufactured grade.

| Parameter                   | Current value                                | Status and source                                                                                                                                                             |
|-----------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $A$                         | 333 s <sup>-1</sup>                          | Value read from <code>phenolic_pyrolysis_native.apdl</code> ; its experimental origin is not included in the project. To be recalibrated using TGA at multiple heating rates. |
| $E_a$                       | 64.081 kJ mol <sup>-1</sup>                  | Same status as $A$ ; the pair $(A, E_a)$ must be identified jointly.                                                                                                          |
| $n$                         | 1                                            | Assumption of a global first-order reaction.                                                                                                                                  |
| $\Delta H_p$                | 418 kJ kg <sup>-1</sup>                      | Endothermic enthalpy encoded in the project; to be replaced by a DSC measurement of the actual resin/fiber system.                                                            |
| $R$                         | 8.314 46 J mol <sup>-1</sup> K <sup>-1</sup> | Universal gas constant.                                                                                                                                                       |
| $\Delta\alpha_{\text{max}}$ | 0.05 per increment                           | Numerical criterion encoded in <code>usermatth.F</code> , not a physical property.                                                                                            |

**Table 4.** Pyrolysis kinetics used in Simulation 1.

The virgin phenolic body `phe1` uses the virgin columns of Table 3 without conversion. The generic epoxy has  $\rho = 1150$  kg m<sup>-3</sup>; its conductivity is 0.25 W m<sup>-1</sup> K<sup>-1</sup> and 0.30 W m<sup>-1</sup> K<sup>-1</sup> at 300 K and 600 K, respectively, and its specific heat capacity is 1000 J kg<sup>-1</sup> K<sup>-1</sup> and 1100 J kg<sup>-1</sup> K<sup>-1</sup> at the same temperatures. The generic aluminum has  $\rho = 2700$  kg m<sup>-3</sup>,  $k = 167$  W m<sup>-1</sup> K<sup>-1</sup>, and  $c_p = 896$  J kg<sup>-1</sup> K<sup>-1</sup>. Within the project, neither material has a complete property curve extending to 2750 K; constant extrapolation is a major risk if heat reaches the outer layers. A qualified study requires the epoxy grade and aluminum alloy to be identified, followed by the addition of their temperature-dependent properties, transitions, decomposition, or melting over the temperature range actually reached.

The char trends—increasing conductivity at high temperature, mass loss, and high heat capacity—are qualitatively consistent with historical charring ablator models [7, 4]. They do not, however, justify the specific numerical values for an unidentified phenolic. NASA uses TGA to measure mass loss and char yield, DSC to determine  $c_p$  and enthalpies, and LFA to determine thermal diffusivity/conductivity [9].

### 2.2.3 MAPDL Code

The following MAPDL block is the file actually associated with the inner material. It creates nine TB,USER points, eleven properties at each temperature, and two state variables:  $\text{SVAR1}=\alpha$  and  $\text{SVAR2}=\dot{\alpha}$ . The `matid` variable is resolved by the Mechanical command context; it must not be hard-coded without first checking the identifier produced by the export.

**Complete MAPDL listing — file `phenolic_pyrolysis_native.apdl`.**

```

1 ! -----
2 ! Prebuilt native Fortran UserMatTh material for inner body phe0
3 ! Solver units: m, kg, s, W, degree C
4 ! -----
5
6 /COM,Prebuilt Fortran UserMatTh pyrolysis model requested
7
8 PYRA = 333.0
9 PYREA = 64081.0
10 PYRN = 1.0
11 PYRDH = 418000.0
12 TO_K = 273.15
13
14 ! TB,USER properties:
15 ! 1 virgin conductivity
16 ! 2 char conductivity
17 ! 3 virgin specific heat
18 ! 4 char specific heat
19 ! 5 virgin density
20 ! 6 char density
21 ! 7 Arrhenius A
22 ! 8 activation energy Ea
23 ! 9 reaction order n
24 ! 10 positive endothermic pyrolysis enthalpy
25 ! 11 Celsius-to-Kelvin offset
26
27 TB,USER,matid,9,11,THERM
28
29 TBTEMP,26.85
30 TBDATA,1,0.25,0.30,900,800,1250,700
31 TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K
32
33 TBTEMP,326.85
34 TBDATA,1,0.32,0.35,1100,900,1200,650
35 TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K
36
37 TBTEMP,626.85
38 TBDATA,1,0.36,0.40,1300,1000,1150,600
39 TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K
40
41 TBTEMP,776.85
42 TBDATA,1,0.36,0.44,1300,1250,1150,600
43 TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K
44
45 TBTEMP,926.85
46 TBDATA,1,0.36,0.49,1300,1550,1150,600
47 TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K
48
49 TBTEMP,1226.85
50 TBDATA,1,0.36,0.68,1300,1950,1150,600
51 TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K
52
53 TBTEMP,1426.85
54 TBDATA,1,0.36,0.81,1300,2060,1150,600
55 TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K
56
57 TBTEMP,1926.85
58 TBDATA,1,0.36,1.24,1300,2085,1150,600
59 TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K
60
61 TBTEMP,2476.85
62 TBDATA,1,0.36,1.73,1300,2090,1150,600
63 TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K
64
65 ! SVAR1 = alpha; SVAR2 = alpha rate
66 TB,STATE,matid,,2
67 TBDATA,1,0.0,0.0

```

### 2.2.4 Fortran UserMatTh Code

The native routine below is the one compiled into `usermatthLib.dll`. It interpolates the TB,USER properties, integrates the kinetics analytically, updates both state variables, and returns  $\mathbf{q} = -k\nabla T$ ,  $\partial u/\partial T = c_p$ , the density, and the specific heat generation  $-\Delta H_p \dot{\alpha}$ . According to the ANSYS documentation, `hgen` is indeed a source per unit mass for purely thermal elements [1].

#### Complete Fortran listing — file `usermatth.F`.

```

1 *deck,usermatth                                USERDISTRIB
2   subroutine usermatth(matId, elemId, kDomIntPt, kLayer,
3   &                    kSectPt, ldstep, isubst, keycut,
4   &                    ncomp, nStatev, nProp, Time, dTime,
5   &                    Temp, dTemp, tgrad, ustatev, prop,
6   &                    coords, dudt, dudg, flux, dfdt,
7   &                    dfdg, cutFactor, hgen, dens,
8   &                    var1, var2, var3, var4, var5, var6)
9 c
10 c   Native UserMatTh model for phenolic pyrolysis.
11 c
12 c   Properties in TB,USER (SI units):
13 c     1   virgin thermal conductivity          [W/(m K)]
14 c     2   char thermal conductivity           [W/(m K)]
15 c     3   virgin specific heat                 [J/(kg K)]
16 c     4   char specific heat                  [J/(kg K)]
17 c     5   virgin density                       [kg/m3]
18 c     6   char density                         [kg/m3]
19 c     7   Arrhenius pre-exponential factor A   [1/s]
20 c     8   activation energy Ea                 [J/mol]
21 c     9   reaction order n                    [-]
22 c    10   positive endothermic enthalpy        [J/kg]
23 c    11   temperature offset                  [K]
24 c
25 c   State variables:
26 c     1   alpha, 0 = virgin and 1 = char
27 c     2   alpha rate                          [1/s]
28 c
29 c   Kinetics:
30 c     d(alpha)/dt = A exp(-Ea/(R T)) (1-alpha)**n
31 c
32 c   The equation is integrated analytically over the current time
33 c   increment. No mutable global state is used, so the routine is
34 c   suitable for shared- and distributed-memory solves.
35 c
36   implicit none
37
38   integer matId, elemId, kDomIntPt, kLayer, kSectPt
39   integer ldstep, isubst, keycut, ncomp, nStatev, nProp
40   integer i, j
41
42   double precision Time, dTime, Temp, dTemp
43   double precision tgrad(ncomp), ustatev(nStatev)
44   double precision prop(nProp), coords(3)
45   double precision dudt, dudg(ncomp), flux(ncomp)
46   double precision dfdt(ncomp), dfdg(ncomp,ncomp)
47   double precision cutFactor, hgen, dens
48   double precision var1, var2, var3, var4, var5, var6
49
50   double precision gasr, minTemp, maxDAlpha, minCut
51   parameter (gasr=8.31446261815324d0)
52   parameter (minTemp=1.0d0)
53   parameter (maxDAlpha=5.0d-2)
54   parameter (minCut=1.0d-1)
55
56   double precision kv, kc, cpv, cpc, rhov, rhoc
57   double precision apre, ea, ordern, deltaH, tempOffset
58   double precision temperatureMid, exponent, rateConstant
59   double precision alphaOld, alphaNew, alphaDot, deltaAlpha
60   double precision betaOld, betaNew, power, bracket
61   double precision conductivity, specificHeat, densityNew
62
63 c   Always initialize every output controlled by UserMatTh.
64   keycut = 0
65   cutFactor = 1.0d0
66   dudt = 1.0d0
67   do i = 1, ncomp
68     dudg(i) = 0.0d0
69     flux(i) = 0.0d0
70     dfdt(i) = 0.0d0
71     do j = 1, ncomp
72       dfdg(i,j) = 0.0d0
73     enddo
74   enddo
75
```

```

76 c      A cutback is safer than accessing an incomplete material table.
77       if (nProp .lt. 11 .or. nStatev .lt. 2) then
78         keycut = 1
79         cutFactor = minCut
80         return
81       endif
82
83       kv      = prop(1)
84       kc      = prop(2)
85       cpv     = prop(3)
86       cpc     = prop(4)
87       rhov    = prop(5)
88       rhoc    = prop(6)
89       apre    = prop(7)
90       ea      = prop(8)
91       ordern  = prop(9)
92       deltaH  = prop(10)
93       tempOffset = prop(11)
94
95 c      Reject nonphysical values without allowing NaNs into the solve.
96       if (kv .le. 0.0d0 .or. kc .le. 0.0d0 .or.
97 &        cpv .le. 0.0d0 .or. cpc .le. 0.0d0 .or.
98 &        rhov .le. 0.0d0 .or. rhoc .le. 0.0d0 .or.
99 &        apre .lt. 0.0d0 .or. ea .lt. 0.0d0 .or.
100 &        ordern .le. 0.0d0 .or. deltaH .lt. 0.0d0) then
101         keycut = 1
102         cutFactor = minCut
103         return
104       endif
105
106       alphaOld = dmax1(0.0d0, dmin1(1.0d0, ustatev(1)))
107       temperatureMid = Temp + 0.5d0*dTemp + tempOffset
108       temperatureMid = dmax1(temperatureMid, minTemp)
109
110       exponent = -ea/(gasr*temperatureMid)
111       exponent = dmax1(-700.0d0, dmin1(700.0d0, exponent))
112       rateConstant = apre*dexp(exponent)
113
114       alphaNew = alphaOld
115       if (dTime .gt. 0.0d0 .and. rateConstant .gt. 0.0d0
116 &        .and. alphaOld .lt. 1.0d0) then
117         betaOld = dmax1(1.0d0-alphaOld, 0.0d0)
118         if (dabs(ordern-1.0d0) .le. 1.0d-10) then
119           betaNew = betaOld*dexp(-rateConstant*dTime)
120         else
121           power = 1.0d0-ordern
122           bracket = betaOld**power
123           - power*rateConstant*dTime
124           if (bracket .le. 0.0d0) then
125             betaNew = 0.0d0
126           else
127             betaNew = bracket**(1.0d0/power)
128           endif
129         endif
130         alphaNew = 1.0d0-betaNew
131         alphaNew = dmax1(alphaOld, dmin1(1.0d0, alphaNew))
132       endif
133
134       deltaAlpha = alphaNew-alphaOld
135       alphaDot = 0.0d0
136       if (dTime .gt. 0.0d0) alphaDot = deltaAlpha/dTime
137
138 c      Linear virgin/char mixture at the updated conversion.
139       conductivity = (1.0d0-alphaNew)*kv + alphaNew*kc
140       specificHeat = (1.0d0-alphaNew)*cpv + alphaNew*cpc
141       densityNew = (1.0d0-alphaNew)*rhov + alphaNew*rhoc
142
143       dudt = specificHeat
144       dens = densityNew
145       do i = 1, ncomp
146         flux(i) = -conductivity*tgrad(i)
147         dfdg(i,i) = -conductivity
148       enddo
149
150       ustatev(1) = alphaNew
151       ustatev(2) = alphaDot
152
153 c      For pure thermal elements hgen is heat generation per unit mass.
154 c      Positive deltaH is endothermic, hence the minus sign.
155       hgen = hgen-deltaH*alphaDot
156
157 c      Limit conversion to five percent per accepted increment.
158       if (deltaAlpha .gt. maxDAlpha) then
159         keycut = 1
160         cutFactor = maxDAlpha/dmax1(deltaAlpha, 1.0d-30)

```



```

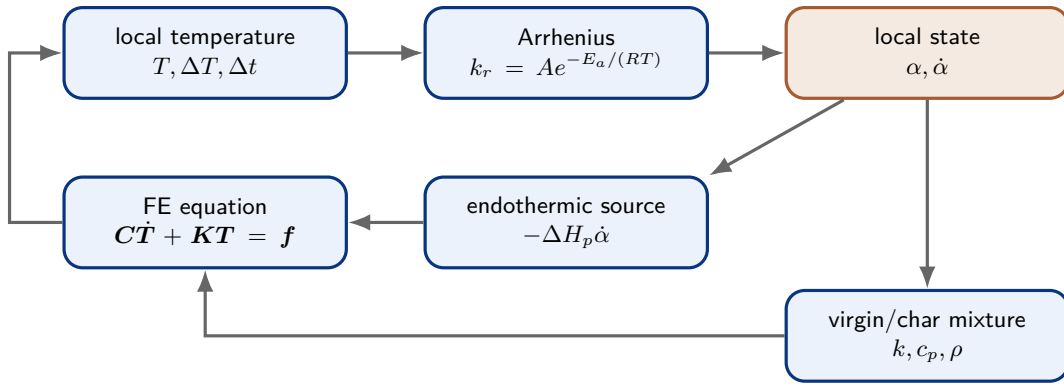
161         cutFactor = dmax1(minCut, dmin1(0.95d0, cutFactor))
162     endif
163
164     return
165 end

```

The Windows x64 build uses Visual Studio 2022 C++ and Intel oneAPI Classic Fortran 2023.1. The `build_usermatth_x64.cmd` script compiles with `/O2 /MD`, links the ANSYS v261 customization libraries, and verifies that the `USERMATTH` symbol is exported. This toolchain must remain aligned with the solver version.

### 2.2.5 Step-by-Step Simulation Preparation

1. **Identify the physical materials.** Record the manufacturer, grade, reinforcement fraction and orientation of the phenolic, the epoxy system, and the aluminum alloy. In the absence of this information, retain an explicit “preliminary data” designation.
2. **Characterize the phenolic.** Perform TGA in an inert atmosphere at several heating rates to obtain  $\rho(T)$ , the char yield, and the kinetics; use DSC for  $c_p(T)$  and  $\Delta H_p$ , LFA for  $k(T)$ , and then measure the apparent density of the char. First fit  $A, E_a, n$  against all TGA curves and validate against a heating rate not used in the fit. A multiple-reaction kinetic model is preferable if a single law cannot reproduce the peaks.
3. **Complete Engineering Data.** Enter the virgin and char tables up to at least the maximum gas temperature, with explicit units. Add the epoxy and aluminum curves over the temperature range actually reached. Do not blindly extrapolate a polymer to 2750 K.
4. **Assign the bodies.** In Mechanical, verify `phe0=PHENOLIC_PYROLYSIS`, `phe1=PHENOLIC_VIRGIN`, `epo=epoxy`, and `alu=aluminum`. Check the assignment in the `ds.dat` export, not solely from the interface color.
5. **Prepare named selections.** Retain the four bodies, the three interface pairs, the hot inner face, and the outer face. Stable names prevent MAPDL commands from being attached to a fragile geometric selection.
6. **Define the contacts.** For this simulation, use bonded thermal contacts and verify that each pair is closed. If an adhesive is meshed as a volume, do not also add a contact resistance that would count the same effect twice.
7. **Mesh and test mesh independence.** Retain at least eight layers through `phe0` for the reference calculation, then compare the wall temperature and the  $\alpha = 0.5$  depth against a finer radial mesh. Mesh independence must be assessed using these observables, not only the maximum temperature.
8. **Build the DLL.** From a properly initialized x64 command prompt, run `build_usermatth_x64.cmd`; check the return code, `compile.log`, `link.log`, and the `USERMATTH` export. Copy the DLL to the solver directory, or otherwise make it accessible there, before starting the run.
9. **Insert the MAPDL block.** Under the thermal-analysis branch, insert a command object that applies the contents of Section 2.2.3 to the `phe0` material. After generating the solver input, verify the presence of `TB,USER`, `TB,STATE`, and nine `TBTEMP` entries.
10. **Define the loading.** Prescribe 295.15 K initially, the hot convection of Equation (19), and then the outer convection and radiation. Ultimately, replace  $T_g(t)$  and  $h(t)$  with profiles from a test or a chamber calculation. Also vary  $h$  over at least 500 to 2000 W m<sup>-2</sup> K<sup>-1</sup> to quantify its influence.
11. **Set time stepping and convergence controls.** Use automatic time stepping between 0.001 s and 0.05 s, retain the  $\Delta\alpha \leq 0.05$  limit, and save `SVAR`. A single 7 s step would eliminate resolution of the front and would not be physically acceptable, even though the local law is integrated analytically.
12. **Verify solver startup.** In the solver output, confirm that the user library is loaded, that the requested number of processes is used, that no `UserMatTh` error occurs, and that the state variables are present. First run a coupon or a very small sector for 0.1 s.
13. **Validate using conservation balances.** For every result set, check  $0 \leq \alpha \leq 1$ , the monotonicity of  $\alpha$  in time, the heat balance, and the integrated volatile mass obtained from Equation (8). Finally, compare a case without pyrolysis and a simplified adiabatic case against an analytical conduction solution.



**Figure 2.** Local coupling in Simulation 1. Conversion modifies both the heat source and the properties at the following substep.

### 2.3 Simulation 1 Results (to be completed)

Subsection reserved for results: temperatures, SVAR1, SVAR2, pyrolysis depth, energy balances, and mesh-independence study.

### 3 Simulation 2: Outgassing and Element Deactivation in Mechanical

#### 3.1 Proposed Theory and Derivation

##### 3.1.1 Solid Mass Balance and Gas Generation

Simulation 2 retains the kinetics used in Simulation 1, but distinguishes the solid residue from the gaseous products. Over an initial volume, the loss of condensed-phase mass is the gas source:

$$\frac{\partial \rho_s}{\partial t} = -\dot{\omega}_g''', \quad \dot{\omega}_g''' = (\rho_v - \rho_c)\dot{\alpha}. \quad (21)$$

The final gas yield of the single-reaction model is therefore

$$Y_g = \frac{\rho_v - \rho_c}{\rho_v} = 1 - \frac{\rho_c}{\rho_v}. \quad (22)$$

With the current tables,  $Y_g$  ranges from 0.44 to 0.48. A nominal value of  $Y_g = 0.47$  is consistent with the model, but must be confirmed using the residual mass measured by TGA.

In a complete porous-ablator model, conservation of gas mass is written as

$$\frac{\partial(\phi \rho_g)}{\partial t} + \nabla \cdot (\rho_g \mathbf{u}_g) = \dot{\omega}_g''', \quad (23)$$

where  $\phi$  is the porosity and  $\mathbf{u}_g$  is the superficial velocity. For slow flow through the char, Darcy's law and the ideal-gas law give

$$\mathbf{u}_g = -\frac{K(\alpha, T)}{\mu_g} \nabla p_g, \quad \rho_g = \frac{p_g M_g}{RT_g}. \quad (24)$$

Ablative-material response codes such as CHAR solve coupled forms of these mass and energy balances [5, 6, 4]. A standard thermal element in Mechanical, however, has no pressure degree of freedom. Consequently, **UserMatTh** cannot solve Equation (23) by itself. Remaining within Mechanical therefore requires either a substantially more complex user-defined multiphysics element or a reduced-order model.

##### 3.1.2 Reduced Radial Closure Compatible with Mechanical

The proposed model assumes that the gases generated in **phe0** escape rapidly toward the hot wall and that their accumulation is negligible. In cylindrical axisymmetry, the gas mass flux reaching the surface at radius  $r_s$  is then obtained directly by integrating the source:

$$\dot{m}_g''(r_s) = \frac{1}{r_s} \int_{r_s}^{R_0} \dot{\omega}_g'''(r) r \, dr, \quad (25)$$

where  $R_0$  is the outer boundary of **phe0**. This expression follows from the balance  $2\pi r_s L \dot{m}_g'' = \int_{r_s}^{R_0} 2\pi r L \dot{\omega}_g''' \, dr$ . It conserves total gas mass even when pressure is not solved explicitly.

The gas carries sensible enthalpy. A minimal local closure adds the following term to the solid energy equation:

$$\dot{q}_{m,g} = -\frac{\dot{\omega}_g'''}{\rho_s} c_{p,g} (T - T_{\text{ref}}). \quad (26)$$

This term must be used only if  $\Delta H_p$  does not already include this enthalpy; otherwise it would be counted twice. A more faithful formulation adds  $-\nabla \cdot (\dot{m}_g'' h_g)$  to the energy balance, as is done in material-response models [4, 5].

##### 3.1.3 Reduction of Convective Heat Flux by Blowing

Outgassing into the chamber produces a blowing flow that opposes heat transfer. For a steady, planar gas film with constant properties, the source-free energy equation is

$$\dot{m}_g'' c_{p,g} \frac{dT}{dy} = k_g \frac{d^2 T}{dy^2}. \quad (27)$$

Integrating between the wall and the outer edge of the film, and expressing the film thickness in terms of the no-blowing coefficient  $h_0 = k_g/\delta$ , gives the classical correction in the form

$$B_h = \frac{\dot{m}_g'' c_{p,g}}{h_0}, \quad F(B_h) = \frac{B_h}{\exp(B_h) - 1}, \quad (28)$$

$$q''_{\text{conv}} = h_0 F(B_h)(T_g - T_s). \quad (29)$$

Because  $F(0) = 1$  and  $F(B_h) < 1$  for  $B_h > 0$ , the model recovers the current convection condition in the absence of outgassing and progressively reduces the heat flux as gas production increases. This law is an engineering closure; it does not replace a boundary-layer calculation and must be calibrated against chamber pressure, propellant composition, and test data.

### 3.1.4 Surface Balance, Recession, and Element-Deactivation Criterion

Pyrolysis produces char; it does not automatically remove that char. Recession requires a second mechanism—oxidation, sublimation, or mechanical erosion—and a surface energy balance:

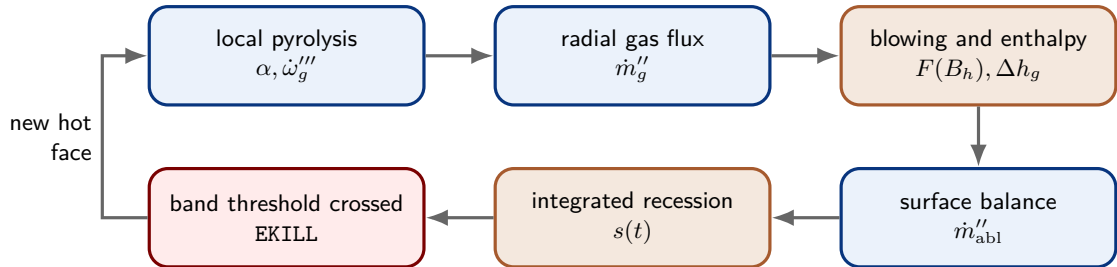
$$q''_{\text{conv}} + q''_{\text{rad}} = q''_{\text{cond}} + \dot{m}_g'' \Delta h_g + \dot{m}_{\text{abl}}'' H_{\text{eff}}. \quad (30)$$

In the absence of sufficient chemical data, the following bounded, energy-controlled law is proposed:

$$\dot{m}_{\text{abl}}'' = \max \left[ 0, \frac{q''_{\text{conv}} + q''_{\text{rad}} - q''_{\text{cond}} - \dot{m}_g'' \Delta h_g}{H_{\text{eff}}} \right], \quad \dot{s} = \frac{\dot{m}_{\text{abl}}''}{\rho_c}. \quad (31)$$

The recession depth is  $s(t) = \int_0^t \dot{s} dt$ . The **phe0** elements are arranged in radial bands; a band is deactivated only when  $s$  exceeds its cumulative thickness. The criterion is therefore the integrated recession, possibly combined with  $\alpha \geq 0.99$ , but never  $\alpha = 1$  alone.

In MAPDL, **EKILL** multiplies an element's contribution by a near-zero factor; the element remains in the mesh. A deactivated thermal element no longer contributes heat capacity or applied loads, and its state variables are reset when it is killed [2, 3]. Any residual mass and energy must therefore be recorded before **EKILL** and included in the balance. After each band is deactivated, the hot-side convection and surface elements must be transferred to the newly exposed face. With only eight layers through **phe0**, recession would advance in increments of approximately 0.318 mm. A credible recession study should use 20 to 40 radial bands, or demonstrate that the existing increment is sufficiently small.



**Figure 3.** Proposed physical sequence for Simulation 2 in Mechanical. Element deactivation is driven by recession, not directly by **SVAR1**.

### 3.1.5 Proposed Initial Values and Sensitivity Studies

Table 5 provides an initial parameter set for prototyping, not for qualification. The permeability values are taken from the **CHAR PICA** model and are only surrogates for a dense phenolic [6]. Phenolic resin products include, among others,  $\text{H}_2$ , water vapor, and aromatic hydrocarbons, and depend on heating rate [8]; representing them with a single pseudo-species is therefore a major simplification.

| Quantity                               | Proposed nominal                        | Sensitivity range                     | Rationale / required measurement                                                                                                                     |
|----------------------------------------|-----------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gas yield $Y_g$                        | 0.47                                    | 0.44–0.50                             | Derived from the current virgin/char densities using Equation (22); replace with TGA data.                                                           |
| Virgin porosity $\phi_v$               | 0.13                                    | 0.05–0.20                             | Derived from $1 - \rho_v / \rho_{\text{skeletal}}$ , with $\rho_{\text{skeletal}} = 1430 \text{ kg m}^{-3}$ , a historical phenolic-nylon value [7]. |
| Char porosity $\phi_c$                 | 0.55                                    | 0.45–0.65                             | Same calculation using $\rho_c = 600 - -700 \text{ kg m}^{-3}$ ; measure by pycnometry/porosimetry.                                                  |
| Virgin permeability $K_v$              | $10^{-12} \text{ m}^2$                  | $10^{-14}$ – $10^{-11} \text{ m}^2$   | PICA “Model A” value used as a surrogate [6]; not transferable without testing.                                                                      |
| Char permeability $K_c$                | $10^{-10} \text{ m}^2$                  | $10^{-12}$ – $10^{-9} \text{ m}^2$    | Same source; logarithmic interpolation with respect to $\alpha$ is recommended.                                                                      |
| Gas viscosity $\mu_g$                  | $4 \times 10^{-5} \text{ Pa s}$         | $2$ – $6 \times 10^{-5} \text{ Pa s}$ | Provisional hot pseudo-gas value; a mixture calculation is required.                                                                                 |
| Molar mass $M_g$                       | $0.024 \text{ kg mol}^{-1}$             | 0.018–0.030                           | Pseudo-species spanning light gases and vapor; calculate from the measured composition.                                                              |
| Gas heat capacity $c_{p,g}$            | $1.5 \text{ kJ kg}^{-1} \text{ K}^{-1}$ | 1.2–2.0                               | Initial pseudo-gas value; use a $c_p(T, p)$ table after characterization.                                                                            |
| Char emissivity $\varepsilon_c$        | 0.80                                    | 0.70–0.90                             | Historical phenolic-nylon starting point [7]; high-temperature measurement required.                                                                 |
| Ablation enthalpy $H_{\text{eff}}$     | $35 \text{ MJ kg}^{-1}$                 | 20, 35, and $50 \text{ MJ kg}^{-1}$   | Effective recession parameter. $50 \text{ MJ kg}^{-1}$ is a historical sublimation anchor [7]; calibrate against thickness loss.                     |
| Reference temperature $T_{\text{ref}}$ | 295.15 K                                | fixed                                 | Consistent with the initial state.                                                                                                                   |
| Recession increment                    | $\dot{s}\Delta t \leq 0.2\Delta r$      | 0.1–0.25 $\Delta r$                   | Numerical criterion to prevent several bands from being crossed in one substep.                                                                      |

**Table 5.** Proposed initial parameters for Simulation 2. They must be treated as uncertain variables until they have been characterized.

Before a complete motor simulation is attempted, three separate validation campaigns are required: (i) TGA/DSC/LFA coupon testing for decomposition, (ii) permeability and gas-pressure measurements in a heated coupon for outgassing, and (iii) an instrumented flame or motor test for  $h_0$ , blowing, and recession. The model must not compensate for an incorrect chamber boundary condition by artificially adjusting  $H_{\text{eff}}$ .

### 3.2 Implementation in Mechanical (to be completed)

Subsection reserved for the future implementation: band architecture, MAPDL commands, load transfer, restart logic, and conservation checks.

### 3.3 Simulation 2 Results (to be completed)

Subsection reserved for the results: outgassing mass flow rate, blowing factor, recession, chronology of deactivated bands, and mass and energy balances.

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